Jun-Kun Chen

Ph.D. Candidate in Computer Science | Expected Dec. 2026

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Legal name: Junkun Chen | Publications: Jun-Kun Chen / J.-K. Chen

Research Focus

Research in **diffusion-based controllable image/video generation**, with first/co-first-author work at NeurIPS and CVPR and internships at Meta and SpreeAI. Specializes in **supervision, training-data design, and curricula** for new generative capabilities, with current work on visual world models and multimodal post-training.

Industry Experience

Meta Reality Labs — **May–Dec. 2025; May–Aug. 2023**
Research Scientist Intern (2025 continued part-time Aug.–Dec.) — Bay Area, CA, U.S. / Zurich, Switzerland

- **2025:** Designed a **GPU-role-separated pipeline** for data processing and model training at **128–256-GPU scale**, supporting 3D/video-generation research on Meta's distributed infrastructure.
- **2023:** Designed and implemented **self-supervised diffusion adaptation** and an asynchronous training pipeline for *ConsistDreamer* (CVPR 2024).

SpreeAI — **May 2024 – May 2025**
AI Research Intern — Remote, U.S.

- **Dress&Dance**: designed **target-steered supervision and staged training** for garment, identity, and motion control in video diffusion, generating 5-second try-on videos at **1152×720, 24 FPS**.
- **Virtual Fitting Room** (NeurIPS 2025): designed **reference-conditioned training from short-clip pairs**, enabling anchored autoregression for identity-consistent video up to **90 seconds**.

Selected Publications

1. Virtual Fitting Room: Generating Arbitrarily Long Videos of Virtual Try-On from a Single Image ↗ — **NeurIPS 2025**
Jun-Kun Chen, Aayush Bansal, Minh Phuoc Vo, Yu-Xiong Wang — Project | Early preview (arXiv, Sep. 2025)

2. Dress&Dance: Dress up and Dance as You Like It ↗ — **WACV 2027**
Jun-Kun Chen, Aayush Bansal, Minh Phuoc Vo, Yu-Xiong Wang — Project | Early preview (arXiv, Aug. 2025)

3. ProEdit: Simple Progression is All You Need for High-Quality 3D Scene Editing ↗ — **NeurIPS 2024**
Jun-Kun Chen, Yu-Xiong Wang — Project | arXiv (Nov. 2024)

4. ConsistDreamer: 3D-Consistent 2D Diffusion for High-Fidelity Scene Editing ↗ — **CVPR 2024**
Jun-Kun Chen, Samuel Rota Bulò, Norman Müller, Lorenzo Porzi, Peter Kontschieder, Yu-Xiong Wang — Project | arXiv (Jun. 2024)

5. Instruct 4D-to-4D: Editing 4D Scenes as Pseudo-3D Scenes Using 2D Diffusion ↗ — **CVPR 2024**
Linzhan Mou*, **Jun-Kun Chen***, Yu-Xiong Wang — Project | arXiv (Jun. 2024)

6. HDEdit: Editing Videos and 3D Scenes with Video Diffusion Through Hierarchical Task Decomposition ↗ — **3DV 2026**
Yanming Zhang*, **Jun-Kun Chen***, Jipeng Lyu, Yu-Xiong Wang — Project | Earlier version (arXiv, Mar. 2025)

*Equal contribution. Earlier publications also appear as Junkun Chen / J. Chen.

Current Research: Generative World Models

Research direction: connecting language reasoning with visual generation, using predicted outcomes as training feedback.

Computer Use — Action-Conditioned Image-Diffusion World Model

- Built a next-screen world model on SenseNova MoT: a language Reasoner describes the action's effects, and a Generator adapted by **diffusion post-training** renders the next screen.
- Formulated **teacher-grounded supervision** for action effects and content preservation, now used to train the Reasoner.

Physics — Video-Diffusion World Model for Physical Events

- Adapted **Cosmos3 video diffusion** to predict physical events from past RGB frames and structured event descriptions.
- Implemented **teacher distillation** from a full-video teacher to a Reasoner that sees only past frames.

Robotics — Motion-Conditioned Video Diffusion World Model

- Trained a **motion-conditioned video world model** on simulated and real robot data (MimicGen, RoboTwin, DROID, RH20T, AgiBot) to predict interaction video from a scene, recent frames, and rendered robot motion.
- Implemented **visual-proxy and paired-correspondence conditioning**; studying adaptation to unseen robot setups.

Education

University of Illinois Urbana-Champaign — **2021 – Dec. 2026 (expected)**
Ph.D. Candidate in Computer Science; Advisor: Prof. **Yu-Xiong Wang** — Urbana, IL

Tsinghua University — **2017 – 2021**
B.Eng. in Computer Science, Yao Class (IIIS); Major GPA: 3.94/4.0 — Beijing, China

Technical Skills & Honors

Tools & Methods: PyTorch; Python; C++; image/video diffusion; diffusion Transformers (DiT); LoRA; vision-language models; multimodal conditioning; training-data construction; benchmark design; quantitative and human evaluation; diffusion post-training.
Competition Highlights: ICPC EC-Final Gold (4th); ICPC Qingdao Gold (2nd); CCPC Harbin Gold; NOI Gold.

Selected Research Contributions

Learning Visual Control and Long-Horizon Generation — Dress&Dance and VFR

- **Dress&Dance:** designed **target-steered supervision and staged curricula** for new visual controls, achieving **20–59% lower FVD** than a Kling 1.6 try-on/animation pipeline on Internet and captured benchmarks (Table C.4).
- Designed multimodal conditioning and **synthetic-condition/real-target triplets**; implemented hybrid image/video training and progressive schedules to add garment and identity control while retaining pretrained priors.
- **Virtual Fitting Room:** designed **short-clip pair sampling** to learn reference-conditioned appearance and prefix continuation, enabling generated 360° anchors at inference for identity-consistent video up to **90 seconds**.

General-Purpose Diffusion Editing — HDEdit and ProEdit

- **HDEdit:** designed **hierarchical task decomposition** to separate instruction following from content preservation, combining LLM scheduling with interpolation refinement for training-free video and scene editing.
- **ProEdit:** designed and implemented tuning-free preservation control and a **progressive editing workflow** with previews and edit-strength control for high-fidelity geometry and appearance editing.

Multi-View and Temporal Consistency — ConsistDreamer and Instruct 4D-to-4D

- **ConsistDreamer:** designed **cross-view self-supervision and staged diffusion adaptation** using warped-view targets, geometry-aware ControlNet/LoRA, and consistency losses across the **denoising trajectory**.
- Implemented an **asynchronous multi-GPU pipeline** that overlaps synchronous diffusion training and image generation with buffered scene fitting on a dedicated GPU.
- **Instruct 4D-to-4D:** extended 2D diffusion to long-term dynamic scenes through **anchor-aware attention** and optical-flow-guided sliding windows.

Geometric Visual Representations — NeuralEditor and PointTree

- Built point-based radiance-field and relaxed k-d-tree representations for editable scene geometry and **transformation-robust point-cloud encoding**.

Additional Publications

7. SceneCraft: Layout-Guided 3D Scene Generation ↗ **NeurIPS 2024**
8. Contrastive Learning Relies More on Spatial Inductive Bias Than Supervised Learning: An Empirical Study ↗ **ICCV 2023**
9. NeuralEditor: Editing Neural Radiance Fields via Manipulating Point Clouds ↗ **CVPR 2023**
10. PointTree: Transformation-Robust Point Cloud Encoder with Relaxed K-D Trees ↗ **ECCV 2022**
11. Is Self-Supervised Contrastive Learning More Robust Than Supervised Learning? ↗ **ICML 2022 Workshop**
12. TorchDrug: A Powerful and Flexible Machine Learning Platform for Drug Discovery ↗ **arXiv preprint, 2022**
13. RNNLogic: Learning Logic Rules for Reasoning on Knowledge Graphs ↗ **ICLR 2021**
14. Pre-training Assisted Scene-aware Dialogue Generation ↗ **DSTC8 Workshop at AAAI 2020**

Earlier Experience

Mila – Quebec AI Institute **Feb. – Aug. 2020**
Research Intern; knowledge-graph reasoning with Prof. Jian Tang Montreal, Canada

Baidu, Natural Language Processing Department **July 2019 – Jan. 2020**
Research / Development Intern; scene-aware dialogue generation (DSTC8 top-3 system) Beijing, China

Competitions, Leadership & Coursework

Programming Competitions: ICPC EC-Final 2018, Gold Medal (4th); ICPC Qingdao Regional 2018, Gold Medal (2nd); CCPC Harbin Regional 2017, Gold Medal; NOI 2016, Gold Medal.
Additional Distinctions: CTSC 2017 (7th); CTSC 2016 Gold (8th); NOI Winter Camp 2016 Gold (3rd).
Leadership & Problem Setting: Problem setter for NOI, CTSC, ICPC, and CCPC events; Vice Chair, Tsinghua Student Association of Algorithms and Competitions (2018–2019).
Selected Coursework (Generative Models): Deep Generative Models (A+); Generative Models for Biomedicine and Life Sciences (A).
Vision & 3D: 3D Vision (A+); Deep Learning for Computer Vision (A); Computer Vision (A); Computational Photography (A+).
Learning: Meta-Learning (A+); Machine Learning (A).